

CS-AML

MVP Engineering Breakdown

Version 0.1.1

Approved Internal Specification Baseline

2026-10-08 · tag v0.1.1-spec

Civil Society Anti-Money Laundering & Financial Intelligence Framework

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Contents

1. Purpose and Engineering Objective	3
1.1 Traceability chain	3
1.2 Engineering principles	3
2. MVP Scope Baseline	3
3. Delivery Strategy and Increments	4
4. Dependency Model	5
5. Detailed Engineering Backlog	5
E0 — Platform Foundation & Delivery	5
E1 — Identity, Access & Protected Sources	6
E2 — Case & Investigation Workflow	7
E3 — Source, Evidence & Document Intake	9
E4 — Entity Resolution & Relationship Model	11
E5 — Timeline & Follow-the-Value	13
E6 — Typology, Hypothesis & Assessment	15
E7 — Search & Investigation Graph	17
E8 — Intelligence Product, Review & Dissemination	17
E9 — Administration, Retention & Operational Readiness	19
6. Cross-Cutting Engineering Requirements	21
6.1 Authorization	21
6.2 Provenance and audit	21
6.3 Data integrity	21
6.4 Testing	21
7. Suggested Repository and Module Boundaries	22
8. Engineering Definition of Ready	22
9. Engineering Definition of Done	22
10. MVP Release Evidence Package	22
11. Post-MVP Deferred Work	23
Annex A — Epic Exit Checklist	23
Annex B — MVP End-to-End Acceptance Scenario	23
Annex C — Initial Engineering Decision Log	24

Engineering execution baseline for MVP 0.1

From SRS requirements to epics, stories, tasks, acceptance tests, dependencies, and release evidence.

Document status — v0.1.1

Version: 0.1.1 — Approved Internal Specification Baseline (2026-10-08, tag v0.1.1-spec). [v0.1.1 · A01]

Supersedes: CS-AML MVP Engineering Breakdown v0.1. The DOCX/PDF files in this repository are the unchanged v0.1 baseline (legacy); this Markdown file is the canonical source.

Validation: approved by the product owner as the internal specification baseline on 2026-10-08 (decision register and release gates in CHANGELOG.md). No implementation test result or independent audit exists yet. Acceptance criteria in this document are targets, not evidence that tests have passed.

CS-AML is not an external standard or certification. References to FATF, Wolfsberg, PPATK, UNODC or other bodies do not imply their endorsement.

Changes in 0.1.1: see CHANGELOG.md at the repository root (audit findings A01–A16).

Status: Approved Internal Specification Baseline (2026-10-08, tag v0.1.1-spec) — baseline for engineering planning and backlog creation [v0.1.1 · A01]

1. Purpose and Engineering Objective

This document translates the CS-AML Product Requirements Document v0.1.1 and Software Requirements Specification v0.1.1 (both draft for review) [v0.1.1 · A01] into an executable MVP engineering plan. It does not redefine product requirements. It establishes engineering work packages, sequencing, acceptance evidence, and a common definition of completion.

MVP release objective: a two-analyst team SHALL complete one sensitive investigation from case opening through independently reviewed and approved intelligence product, with provenance, authorization, and auditability intact.

1.1 Traceability chain

Framework → Product Feature → PRD → SRS → Epic → Story → Engineering Task → Acceptance Test → Release Evidence

1.2 Engineering principles

- Vertical slices over component completion.
- Canonical data first; search and graph remain rebuildable projections.
- Secure-by-default authorization is implemented in services/API, not only presentation logic.
- Analytical uncertainty is preserved in schema, UX, export, and tests.
- Every critical capability is accepted through observable behavior, not merely code presence.
- MVP integration scope is deliberately narrow; advanced OCR, sanctions, blockchain analytics, AI assistant and federation remain post-MVP unless required to unblock the release scenario.

2. MVP Scope Baseline

The P0 baseline is the mandatory release scope defined by the PRD/SRS. The engineering backlog groups those requirements into ten epics and eight delivery increments. P1/P2 capabilities are excluded unless explicitly promoted through change control.

Epic	Name	Release outcome	Primary traceability
E0	Platform Foundation & Delivery	Establish the deployable application skeleton, environments, CI/CD, canonical database, evidence storage, audit substrate, and engineering conventions.	SRS-OPS; SRS-DR; SRS-AUD; F-OPS-001; F-AUD-001
E1	Identity, Access & Protected Sources	Implement OIDC/MFA integration, role/object access controls, case membership, classification, and protected-source compartment.	SRS-SEC; F-SEC-001..003
E2	Case & Investigation Workflow	Implement case register, investigation charter, lifecycle gates, tasks, assignments, and case activity history.	SRS-FR-CASE; F-CASE-001..005
E3	Source, Evidence & Document Intake	Implement source provenance, original evidence preservation, hashing, extracts, derivative lineage, ratings, claim and fact lifecycle, and document ingestion.	SRS-FR-EVD/DOC/CLM; F-EVD-001..006; F-EVD-008; F-DOC-001 [v0.1.1 · A10]
E4	Entity Resolution & Relationship Model	Implement entity registry, aliases/identifiers, candidate matching, merge/unmerge, relationship, ownership/control, and asset registry.	SRS-FR-ENT/REL/AST; F-ENT-001..005; F-REL-001..003; F-AST-001
E5	Timeline & Follow-the-Value	Implement events, visual timeline, ValueFlow records, multi-leg chains, uncertainty classes, range/unknown values, and flow visualization.	SRS-FR-TIM/VAL; F-TIM-001..002; F-VAL-001..004
E6	Typology, Hypothesis & Assessment	Implement typology catalogue, indicators, typology worksheet, competing hypotheses, gaps, confidence, disconfirming-search record, and assessments.	SRS-FR-TYP/HYP/ASM (incl. SRS-FR-ASM-004); F-TYP-001..003; F-HYP-001..003; F-ASM-001..003 [v0.1.1 · A05]
E7	Search & Investigation Graph	Implement permission-aware full-text/faceted search and evidence-backed graph visualization with canonical links.	SRS-FR-SCH/GRF; F-SCH-001..002; F-GRF-001
E8	Intelligence Product, Review & Dissemination	Implement report templates, evidence index, peer review, versioning/corrections, approvals, export/referral, and dissemination log.	SRS-FR-PRD/REV/DIS; F-PRD-001..003; F-REV-001; F-DIS-001..004
E9	Administration, Retention & Operational Readiness	Implement vocabularies, retention policies, backup/restore, configuration, minimum monitoring, security hardening, and release verification.	SRS-FR-ADM/SEC/AUD/OPS; F-ADM-001; F-ADM-003; F-OPS-001

3. Delivery Strategy and Increments

Increments, sequencing and any sizing derived from this backlog are planning targets, not measured estimates or commitments; no task estimates, velocity data or benchmarks exist yet. [v0.1.1 · N06]

The recommended sequence creates demonstrable end-to-end capabilities early. Increments are dependency-aware, but teams MAY overlap work where contracts are stable and integration tests remain authoritative.

Increment	Primary scope	Exit outcome
I0 — Engineering Foundation	E0 + core E1	Runnable secure skeleton; migrations, canonical DB, object storage, audit, OIDC baseline.
I1 — Governed Case Workspace	E1 + E2	Authenticated team can create compartmentalized case, charter, gates, tasks and activity.
I2 — Evidence-to-Entity Chain	E3 + E4 partial	Team can register source, preserve evidence, cite extracts and build reviewable entities.
I3 — Investigation Model	E4 remainder + E5	Relationships, assets, timeline and value-flow analysis become usable end-to-end.
I4 — Analytical Reasoning	E6	Typology, hypotheses, gaps, assessment and confidence complete analytical chain.
I5 — Discovery & Graph	E7	Search and graph improve navigation without becoming canonical truth.
I6 — Reviewed Intelligence Product	E8	Assessment becomes reviewed/versioned intelligence product with controlled export.
I7 — Operational Release	E9 + regression	Retention, restore, hardening and full MVP release scenario verified.

4. Dependency Model

Core dependency direction: E0 precedes all epics; E1 is required before sensitive data becomes multi-user; E2/E3 establish governed investigation context and evidence; E4/E5 create the financial investigation model; E6 creates analytical reasoning; E7 accelerates discovery; E8 creates controlled intelligence outputs; E9 closes operational release risk.

Critical path: E0 → E1 → E2 → E3 → E4 → E5 → E6 → E8 → E9. E7 can begin after E4 canonical relationships are stable.

5. Detailed Engineering Backlog

E0 — Platform Foundation & Delivery

Establish the deployable application skeleton, environments, CI/CD, canonical database, evidence storage, audit substrate, and engineering conventions.

Traceability: SRS-OPS; SRS-DR; SRS-AUD; F-OPS-001; F-AUD-001

ST-E0-01 — Repository and service skeleton

As an engineer, I need a reproducible project structure so all later capabilities share conventions.

Engineering tasks

- Create backend, frontend, worker and infrastructure directories
- Configure environment-based settings and secrets
- Define coding/lint/test conventions
- Add health endpoint and build metadata

Acceptance tests / exit criteria

- Fresh checkout can run locally from documented commands
- CI executes lint + unit tests
- No production secret is committed

ST-E0-02 — Canonical PostgreSQL baseline

As a data steward, I need versioned canonical storage so analytical objects have a durable source of truth.

Engineering tasks

- Create database migration framework
- Implement common object envelope: UUID, timestamps, version, classification, created_by
- Add soft/superseded state conventions
- Create migration rollback/test strategy

Acceptance tests / exit criteria

- Migrations apply to empty database
- Schema version is queryable
- Canonical IDs remain stable across updates

ST-E0-03 — Evidence object storage baseline

As an evidence custodian, I need originals stored separately from metadata.

Engineering tasks

- Configure S3-compatible or filesystem evidence store
- Use opaque object keys, not user filenames
- Store checksum and content metadata in DB
- Block in-place overwrite of originals

Acceptance tests / exit criteria

- Uploading same filename twice creates distinct evidence versions
- Original object cannot be modified by analyst role

ST-E0-04 — Audit event substrate

As an auditor, I need material changes captured consistently.

Engineering tasks

- Define AuditEvent schema
- Create application audit helper/middleware
- Capture actor, action, object, before/after reference, timestamp, request correlation
- Protect audit records from ordinary modification

Acceptance tests / exit criteria

- Create/update/delete/merge/export actions produce events
- Analyst cannot alter audit history

E1 — Identity, Access & Protected Sources

Implement OIDC/MFA integration, role/object access controls, case membership, classification, and protected-source compartment.

Traceability: SRS-SEC; F-SEC-001..003

ST-E1-01 — OIDC login and session policy

As a user, I need organisational SSO so access can be centrally revoked.

Engineering tasks

- Implement OIDC authorization code flow (+ PKCE) with Django as confidential client of Keycloak, server-side session (BFF): GET /auth/login, GET /auth/callback, POST /auth/logout, GET /auth/session [v0.1.1 · A11]
- Browser holds only __Host-csam1_session cookie (HttpOnly, Secure, SameSite=Lax, Path=/); access/refresh/ID tokens stay server-side; no Authorization: Bearer from browser; CSRF token in X-CSRFToken on unsafe methods [v0.1.1 · A11]
- Map IdP subject to local user
- Enforce session timeout and logout (idle/absolute timeouts per security-policy configuration; back-channel logout revokes sessions where supported)
- Support MFA claim/policy checks

Acceptance tests / exit criteria

- Disabled IdP user cannot establish new session

- Session shows authenticated identity and roles
- No OIDC token is readable from browser JavaScript; unsafe request without valid CSRF token is rejected [v0.1.1 · A11]

ST-E1-02 — Role and case membership authorization

As a case owner, I need need-to-know access so sensitive cases stay compartmentalised.

Engineering tasks

- Define roles: investigator, case_owner, reviewer, data_steward, evidence_custodian, admin, auditor
- Implement case membership ACL
- Apply authorization in API layer, not UI only
- Add classification checks

Acceptance tests / exit criteria

- Direct API request cannot bypass case membership
- Unauthorized object IDs return non-disclosing denial

ST-E1-03 — Protected-source compartment

As a source handler, I need source identity separated from routine evidence.

Engineering tasks

- Create ProtectedSource model/store
- Separate identity from source-derived evidence reference
- Restrict handler role
- Redact identity from normal search/index/audit payloads

Acceptance tests / exit criteria

- Routine analyst can use source-derived evidence without source identity
- Protected identity never appears in normal export

E2 — Case & Investigation Workflow

Implement case register, investigation charter, lifecycle gates, tasks, assignments, and case activity history.

Traceability: SRS-FR-CASE; F-CASE-001..005

ST-E2-01 — Case register and creation

As an investigator, I need a stable case workspace.

Engineering tasks

- Create Case model/API/UI
- Required fields: title, purpose, owner, classification, status
- Generate stable case identifier
- Case list filters by status/owner/classification

Acceptance tests / exit criteria

- Case cannot activate without owner, purpose, question
- Case history records creation

ST-E2-02 — Investigation Charter

As a case owner, I need scope and boundaries recorded before substantive work.

Engineering tasks

- Implement investigation question, scope included/excluded, jurisdictions, period, risks, authorised/prohibited collection
- Version charter
- Require reason for material scope change

Acceptance tests / exit criteria

- Old charter version remains retrievable
- Scope change produces audit event

ST-E2-03 — Lifecycle gates G0-G6

As governance reviewer, I need high-risk actions controlled by gates.

Engineering tasks

- Model gates and gate requirements
- Gate submission/approval/rejection
- Prevent self-approval when policy requires independence
- Record conditions and unresolved gaps

Acceptance tests / exit criteria

- Controlled transition is blocked without approval
- Approval predates downstream action

ST-E2-04 — Tasks and assignments

As an investigator, I need actionable work tracked.

Engineering tasks

- Task model with owner/status/due date/type
- Case task board/list
- Completion history
- Link tasks to sources/entities/hypotheses where relevant

Acceptance tests / exit criteria

- Task ownership and completion are visible
- Closed task retains historical state

ST-E2-05 — Case activity timeline

As a reviewer, I need to reconstruct material case activity.

Engineering tasks

- Project relevant AuditEvents into case activity
- Filter by actor/action/object
- Link activity to canonical object

Acceptance tests / exit criteria

- Material case events appear chronologically

- Activity item opens the referenced object when authorized

E3 — Source, Evidence & Document Intake

Implement source provenance, original evidence preservation, hashing, extracts, derivative lineage, ratings, claim and fact lifecycle, and document ingestion.

Traceability: SRS-FR-EVD/DOC/CLM; F-EVD-001..006; F-EVD-008; F-DOC-001 [v0.1.1 · A10]

ST-E3-01 — Source register

As an investigator, I need material sources registered with provenance.

Engineering tasks

- Source model/API/UI
- Capture origin, publisher, URL/location, publication/access dates, access method, legal note
- Archive reference and reliability fields

Acceptance tests / exit criteria

- Assessment-linked source must contain provenance minimum
- Mutable web source can reference preserved copy

ST-E3-02 — Original evidence ingestion

As a custodian, I need evidence originals preserved.

Engineering tasks

- Upload API with streaming size limits
- Virus/malware scanning hook
- Write immutable original
- Capture MIME/size/filename/acquired_at/collector

Acceptance tests / exit criteria

- Failed ingest never creates misleading completed record
- Original remains unchanged after analyst actions

ST-E3-03 — Hash/integrity verification

As a reviewer, I need to verify evidence integrity.

Engineering tasks

- Compute SHA-256 at ingest
- Store algorithm/hash
- Add reverify action
- Flag mismatch

Acceptance tests / exit criteria

- Recomputed hash matches stored original in normal case
- Mismatch creates security/audit event

ST-E3-04 — Evidence extracts and citations

As an analyst, I need precise excerpts linked to originals.

Engineering tasks

- EvidenceExtract object
- Support page/paragraph/region/location descriptor
- Preview parent context
- Link extract to claims as supporting evidence; facts are created only through the claim and fact lifecycle (ST-E3-07), never directly from an extract [v0.1.1 · A10]

Acceptance tests / exit criteria

- Extract cannot exist without parent evidence
- Citation opens exact parent location where supported
- Creating or linking an extract never creates a Fact [v0.1.1 · A10]

ST-E3-05 — Derivative lineage

As an analyst, I need OCR/translation/processed copies distinguished from originals.

Engineering tasks

- Derivative model and parent link
- Transformation type/tool/version/time/creator
- Prevent derivative from replacing original
- Lineage UI

Acceptance tests / exit criteria

- Every derivative traces to original
- Original and derivative are visually distinct

ST-E3-06 — Reliability and credibility ratings

As an analyst, I need source reliability separate from information credibility.

Engineering tasks

- Implement A-F source scale
- Implement 1-6 information credibility scale
- Require rationale for material ratings
- Allow reassessment history

Acceptance tests / exit criteria

- UI does not merge the two ratings
- Rating change is versioned/audited

ST-E3-07 — Claim and fact lifecycle [v0.1.1 · A10]

As an analyst, I need what a source asserted kept separate from what the team has verified, so that facts are created only through reviewable decisions and corrections reach dependent analysis.

Added in v0.1.1 (audit finding A10). Approved by product owner, 2026-10-08. A claim is never converted into a fact; a fact is a separate object supported by evidence (mandatory), optionally by claims, and by verification decisions. [v0.1.1 · C02]

Engineering tasks

- Claim model/API/UI: source-attributed assertion linked to source/person and evidence/extract refs; `claim_status` (RECORDED, UNDER_REVIEW, CORROBORATED, CONTRADICTED, UNRESOLVED); `disputed` boolean kept for compatibility and derived from `status/open dispute`
- VerificationDecision model: `target_ref` (claim or fact), `decision`, `rationale`, `evidence_refs`, `decided_by`, `decided_at`, optional `review_ref`; append-only
- Fact creation command: requires `supporting_evidence` (1..n) and `decision_rationale`; `supporting_claim_refs` optional; creates the Fact (PROVISIONAL) and its CREATE VerificationDecision atomically in one transaction; `verification_decision_refs` is server-populated; supporting claims are not modified [v0.1.1 · A10] [v0.1.1 · C01] [v0.1.1 · C02]
- Fact transitions: `establish` (Reviewer other than proposer), `dispute` (any authorized case member, with evidence), `supersede` (requires `superseded_by` replacement fact); each command writes its ESTABLISH/DISPUTE/SUPERSEDE decision atomically with the status change [v0.1.1 · C01]
- Dependency tracking: on DISPUTED/SUPERSEDED, flag dependent Assessments and IntelligenceProducts `review_required` with link to the triggering decision; create a correction review task for published products instead of mutating them
- Endpoints per API Specification v0.1.1 (claims, verification-decisions, facts, `establish/dispute/supersede`, dependents); audit events for every transition

Acceptance tests / exit criteria

- Analyst conclusion never overwrites claim content or attribution
- Fact creation without evidence refs or decision rationale is rejected (422); a successful creation returns the fact with its CREATE decision in `verification_decision_refs`, and no fact exists without that decision [v0.1.1 · C01] [v0.1.1 · C02]
- Proposer cannot establish own fact; independent reviewer can
- Verification decisions cannot be edited or deleted
- Pilot path: claim recorded → fact created (PROVISIONAL) supported by evidence and the claim → ESTABLISHED → DISPUTED → SUPERSEDED, each with its decision, preserving full history. Dependent flagging of assessments and products is verified as a regression in ST-E6-06 (Sprint 5) and ST-E8-04 (Sprint 7), when those objects exist [v0.1.1 · C11]

Traceability: SRS-FR-CLM-001..004; F-EVD-008

E4 — Entity Resolution & Relationship Model

Implement entity registry, aliases/identifiers, candidate matching, merge/unmerge, relationship, ownership/control, and asset registry.

Traceability: SRS-FR-ENT/REL/AST; F-ENT-001..005; F-REL-001..003; F-AST-001

ST-E4-01 — Entity registry

As an analyst, I need reusable entities outside case silos.

Engineering tasks

- Entity model by type
- Canonical name, aliases, identifiers, jurisdictions, status
- Case-to-entity association without duplicating entity

Acceptance tests / exit criteria

- Same canonical entity can appear in multiple authorized cases
- Entity record preserves provenance-bearing assertions

ST-E4-02 — Aliases and identifiers

As an analyst, I need conflicting identifiers represented without destructive overwrite.

Engineering tasks

- Identifier records with type/value/source/status
- Alias records with source/time
- Uniqueness rules only where justified

Acceptance tests / exit criteria

- Conflicting values coexist with provenance
- No silent overwrite

ST-E4-03 — Candidate entity matching

As an analyst, I need a reviewable duplicate-candidate screen.

Engineering tasks

- Candidate matching on normalized name + identifiers
- Display matching and conflicting attributes
- No auto-merge in MVP
- Record POSSIBLE_MATCH, KEEP_SEPARATE and DEFER outcomes as ResolutionDecision records; a KEEP_SEPARATE pair is not re-suggested unless new evidence is attached [v0.1.1 · ER]

Acceptance tests / exit criteria

- Candidate screen explains why pair was suggested
- Common name alone does not force merge

ST-E4-04 — Merge and unmerge

As a data steward, I need reversible entity resolution.

Engineering tasks

- ResolutionDecision record (append-only; MERGE with surviving_entity_ref, UNMERGE with reverses_decision_ref; evidence, confidence, rationale, reviewer for high-impact decisions) per Data Model v0.1.1 §8.4 [v0.1.1 · ER]
- Derive entity resolution_status only from ResolutionDecisions (MERGE → MERGED/RESOLVED; UNMERGE → SPLIT) [v0.1.1 · ER]
- Preserve old IDs as aliases/superseded records
- Repoint relationships with history
- Implement unmerge recovery
- Merge, unmerge, POST /resolution-decisions and match-candidate decisions carry body expected_versions (no If-Match); merge/unmerge require Idempotency-Key (API Specification §10, §11, §14) [v0.1.1 · C03]

Acceptance tests / exit criteria

- Unmerge restores prior object topology
- Merge rationale/evidence is mandatory
- Every merge and unmerge produces a ResolutionDecision; a direct resolution_status change without a decision is rejected (SRS-FR-ENT-006) [v0.1.1 · ER]

- Missing or incomplete expected_versions → 428; a stale entry → 412 with details.current_record_versions; a retried merge with the same Idempotency-Key replays the original result [v0.1.1 · C03]

ST-E4-05 — Relationships, ownership and control

As an analyst, I need first-class connections with evidence.

Engineering tasks

- Relationship model with type/endpoints/evidence/confidence/time
- OwnershipInterest with percentage nullable
- ControlAssertion distinguishes legal owner/beneficial owner/controller/user

Acceptance tests / exit criteria

- Graph edge always traces to relationship record
- Unknown percentage remains null, not zero

ST-E4-06 — Asset registry

As an analyst, I need assets linked to ownership/control assertions.

Engineering tasks

- Asset model/types/identifiers/location
- Attribution relationship to person/org
- Evidence and status

Acceptance tests / exit criteria

- Asset attribution distinguishes ownership/control/use
- Material attribution requires evidence link

E5 — Timeline & Follow-the-Value

Implement events, visual timeline, ValueFlow records, multi-leg chains, uncertainty classes, range/unknown values, and flow visualization.

Traceability: SRS-FR-TIM/VAL; F-TIM-001..002; F-VAL-001..004

ST-E5-01 — Events and temporal precision

As an analyst, I need chronology with uncertainty.

Engineering tasks

- Event object with date precision
- Link entities/assets/evidence
- Support exact date, month, year, range, unknown

Acceptance tests / exit criteria

- Approximate date is not rendered as exact
- Event links to supporting evidence

ST-E5-02 — Visual timeline

As an analyst, I need case chronology to reveal patterns.

Engineering tasks

- Timeline query/API
- Filters by entity/type/date
- Interactive UI and export representation

Acceptance tests / exit criteria

- Timeline respects object permissions
- Each item opens canonical event/evidence

ST-E5-03 — ValueFlow canonical model

As an analyst, I need transfer of economic value represented without pretending all flows are transactions.

Engineering tasks

- ValueFlow fields origin/destination/mechanism/value/currency/date/evidence
- Mandatory flow_class enum
- Confidence/status

Acceptance tests / exit criteria

- Cannot save flow without class
- Unknown value allowed without zero substitution

ST-E5-04 — Multi-leg flow builder

As an analyst, I need reconstructable chains.

Engineering tasks

- ValueFlowLeg model/order
- Builder UI
- Per-leg evidence/confidence/class
- Chain validation

Acceptance tests / exit criteria

- Each leg can carry different evidence/class
- Removing a leg does not destroy evidence objects

ST-E5-05 — Flow visualisation

As a reviewer, I need direct vs reconstructed flows visually unmistakable.

Engineering tasks

- Distinct labels/line semantics for four classes
- Legend always visible
- Export keeps class labels

Acceptance tests / exit criteria

- Screenshot/export can distinguish all classes without color alone
- No derived flow shown as direct

E6 — Typology, Hypothesis & Assessment

Implement typology catalogue, indicators, typology worksheet, competing hypotheses, gaps, confidence, and assessments.

Traceability: SRS-FR-TYP/HYP/ASM (incl. SRS-FR-ASM-004, ST-E6-06); F-TYP-001..003; F-HYP-001..003; F-ASM-001..003 [v0.1.1 · C10]

ST-E6-01 — Typology catalogue browser

As an analyst, I need versioned CS-AML typologies in the workflow.

Engineering tasks

- Load catalogue data
- Browse/search typologies
- Display mechanism, observables, false positives, version

Acceptance tests / exit criteria

- Assessment records catalogue version used
- Typology content cannot silently mutate historical case

ST-E6-02 — Indicator capture

As an analyst, I need evidence-linked indicators and counter-indicators.

Engineering tasks

- Indicator model with direction/type/significance
- Links to entities/events/flows/evidence
- Counter-indicator support

Acceptance tests / exit criteria

- Material indicator has evidence/proposition
- Indicator itself is not labelled proof

ST-E6-03 — Typology match worksheet

As an analyst, I need structured comparison rather than automatic accusation.

Engineering tasks

- TypologyMatch object
- Observed indicators/counter-indicators
- Consistency level
- Alternative explanations

Acceptance tests / exit criteria

- Single weak indicator cannot produce strong level automatically
- Analyst rationale required

ST-E6-04 — Competing hypotheses

As an analyst, I need multiple explanations tested.

Engineering tasks

- Hypothesis object/status

- Supporting/contradicting evidence links
- Alternative hypothesis set
- Assumptions

Acceptance tests / exit criteria

- At least two plausible hypotheses supported in pilot
- Rejected hypothesis remains in history

ST-E6-05 — Intelligence gaps

As an analyst, I need unknowns explicit.

Engineering tasks

- Gap object/question/impact/priority/status
- Link to hypothesis/assessment
- Closure rationale

Acceptance tests / exit criteria

- Assessment can list unresolved gaps
- Gap cannot be silently deleted

ST-E6-06 — Assessment and confidence

As an analyst, I need judgement with basis and uncertainty.

Engineering tasks

- Assessment versioning
- Judgement, confidence, rationale, assumptions, gaps, alternatives
- Confidence wire enum HIGH, MODERATE, LOW, INSUFFICIENT_BASIS with mandatory rationale; INSUFFICIENT_BASIS never converted to LOW/null/zero; null only on drafts [v0.1.1 · A09]
- Evidence traversal
- Disconfirming-search record (what was searched, sources consulted, result, rationale) required before a high-impact/adverse assessment or product can pass review (SRS-FR-ASM-004) [v0.1.1 · A05]: Assessment.disconfirming_searches[] (Data Model §13.3) recorded through POST /assessments/{assessmentId}/disconfirming-searches (If-Match on the assessment); POST /reviews/{reviewId}/approve returns 409 STATE_CONFLICT with details. reason = "DISCONFIRMATION_REQUIRED" when the entry is missing [v0.1.1 · C10]

Acceptance tests / exit criteria

- Reviewer can trace assessment backward to evidence
- Confidence basis is mandatory for material assessment
- Review of a high-impact adverse assessment without a disconfirmation record is rejected; after the record is added, review can proceed (tested separately from backward traceability) [v0.1.1 · A05] — rejection happens at review approval (409, DISCONFIRMATION_REQUIRED), not at finalization [v0.1.1 · C10]
- Regression of ST-E3-07: disputing or superseding a supporting fact flags the dependent assessment review_required with review_trigger_ref [v0.1.1 · C11]

Traceability: SRS-FR-ASM-001..004; F-ASM-001..003; SRS-FR-CLM-004 (regression) [v0.1.1 · C10]

- Finalizing an assessment with a null confidence level is rejected; INSUFFICIENT_BASIS round-trips unchanged through DB/API/UI/export [v0.1.1 · A09]

E7 — Search & Investigation Graph

Implement permission-aware full-text/faceted search and evidence-backed graph visualization with canonical links.

Traceability: SRS-FR-SCH/GRF; F-SCH-001..002; F-GRF-001

ST-E7-01 — Permission-aware global search

As an analyst, I need to find authorized entities, cases and evidence.

Engineering tasks

- Search service/index or DB FTS
- Permission filtering before result display
- Filters by object type/case/date

Acceptance tests / exit criteria

- Unauthorized object never appears in count/snippet
- Search result links canonical object

ST-E7-02 — Faceted case search

As an analyst, I need focused search within a case.

Engineering tasks

- Facets for source/entity/event/evidence/relationship
- Case scope enforcement
- Saved query not required in MVP

Acceptance tests / exit criteria

- Facets respect ACL and classification
- Query response time meets MVP NFR target

ST-E7-03 — Investigation graph

As an analyst, I need visual entity/relationship/asset/value-flow exploration.

Engineering tasks

- Graph projection from canonical data
- Node/edge filtering
- Evidence/provenance side panel
- Save view state optionally

Acceptance tests / exit criteria

- Every material edge links to canonical relationship/evidence
- Projection can be rebuilt without data loss

E8 — Intelligence Product, Review & Dissemination

Implement report templates, evidence index, peer review, versioning/corrections, approvals, export/referral, and dissemination log.

Traceability: SRS-FR-PRD/REV/DIS; F-PRD-001..003; F-REV-001; F-DIS-001..004

ST-E8-01 — Intelligence product templates

As an analyst, I need consistent report outputs.

Engineering tasks

- Product model
- Templates (all six MVP P0 templates): Financial Intelligence Note, Entity Profile, Asset Profile, Network Analysis, Referral Package, Case Report [v0.1.1 · A06]
- Template-specific section structure and canonical-object bindings for each of the six templates [v0.1.1 · A06]
- Auto-populate metadata/classification/evidence index references

Acceptance tests / exit criteria

- Draft is versioned
- Product shows author/reviewer/status
- Each of the six templates can be generated from canonical objects with version, classification, author/reviewer and evidence index (SRS-FR-PRD-001) [v0.1.1 · A06]

ST-E8-02 — Evidence index generation

As a reviewer, I need claims traceable from report to evidence.

Engineering tasks

- Generate index from linked facts/assessments, following each fact through its verification decision(s) and source claims/evidence (facts exist only via the ST-E3-07 lifecycle) [v0.1.1 · A10]
- Include source/evidence IDs and citation locations
- Permission-aware inclusion

Acceptance tests / exit criteria

- Key fact in pilot has traversable evidence reference
- Restricted evidence is handled according to export policy
- Index shows fact status; a product that depends on a DISPUTED or SUPERSEDED fact is flagged review_required [v0.1.1 · A10]

ST-E8-03 — Peer review workflow

As a reviewer, I need independent approval and change requests.

Engineering tasks

- Review states/commenting/request-changes/approve
- Prevent author self-approval where required
- Freeze reviewed version

Acceptance tests / exit criteria

- Approved version identifies independent reviewer
- Post-approval change creates new version

ST-E8-04 — Corrections and supersession

As a case owner, I need prior products retained when corrected.

Engineering tasks

- Supersedes relationship
- Status current/superseded/retracted
- Correction note

Acceptance tests / exit criteria

- Old version remains auditable
- Recipient-facing export identifies current version
- Regression of ST-E3-07: disputing or superseding a fact flags dependent products review required, leaves a published product unchanged and creates a correction review task [v0.1.1 · C11]

ST-E8-05 — Dissemination approval and export

As a case owner, I need controlled external release.

Engineering tasks

- Dissemination object recipient/purpose/classification/approval
- Export minimization selector
- Generate package/report

Acceptance tests / exit criteria

- External export blocked before approval
- Export manifest lists included objects/version

ST-E8-06 — Referral package and sharing log

As an investigator, I need structured referral plus immutable sharing record.

Engineering tasks

- Referral template sections
- Sharing log record recipient/time/version/restrictions
- Audit export event

Acceptance tests / exit criteria

- Referral separates fact/analysis/gaps
- Ordinary analyst cannot edit completed sharing log

E9 — Administration, Retention & Operational Readiness

Implement vocabularies, retention policies, backup/restore, configuration, minimum monitoring, security hardening, and release verification.

Traceability: SRS-FR-ADM/SEC/AUD/OPS; F-ADM-001; F-ADM-003; F-OPS-001

ST-E9-01 — Controlled vocabulary admin

As an admin, I need versioned vocabularies.

Engineering tasks

- Relationship/asset/status/classification vocabulary UI
- Version terms
- Disable without deleting historical semantics

Acceptance tests / exit criteria

- Historical record displays original term/version
- Vocabulary change audited

ST-E9-02 — Retention policy engine

As a privacy/admin role, I need retention/disposition states.

Engineering tasks

- RetentionPolicy model
- Apply by classification/object type
- Preview eligible records
- Hold/exception support

Acceptance tests / exit criteria

- Disposition requires authorization
- Legal hold blocks deletion

ST-E9-03 — Backup and restore

As an operator, I need recoverable canonical data and evidence.

Engineering tasks

- Automated DB backup
- Evidence store backup strategy
- Configuration backup
- Restore runbook and test

Acceptance tests / exit criteria

- Documented restore test succeeds
- Restored hashes/evidence references validate

ST-E9-04 — Operational hardening

As an operator, I need production-safe defaults.

Engineering tasks

- Security headers/TLS/reverse proxy
- Secrets management
- Rate limits/upload limits
- Error handling without sensitive leakage
- Minimal health/metrics

Acceptance tests / exit criteria

- No debug mode in production
- Sensitive evidence content absent from routine logs

ST-E9-05 — MVP end-to-end release test

As product owner, I need objective release evidence.

Engineering tasks

- Create representative pilot dataset
- Automate what can be automated
- Run two-analyst scenario
- Record failures and remediation

Acceptance tests / exit criteria

- All mandatory P0 SRS requirements pass or accepted exception exists; no exception is possible for a non-waivable invariant (authorization, source identity, evidence integrity/provenance, certainty promotion, approval bypass, audit history) — see Sprint & Milestone Plan v0.1.1 §7.2 [v0.1.1 · A16]
- Pilot completes case-to-approved-product with intact provenance/audit

6. Cross-Cutting Engineering Requirements

6.1 Authorization

- Every API read/write decision is evaluated server-side.
- Object-level classification and case membership are evaluated before data projection or export.
- Search/graph must not leak inaccessible object existence through counts, snippets, or topology.

6.2 Provenance and audit

- All canonical analytical objects expose stable IDs.
- Material mutations record actor/time/reason and relevant before/after references.
- Evidence extracts, derivatives, facts, indicators, hypotheses and assessments remain traversable backward to sources.

6.3 Data integrity

- Original evidence cannot be edited in place.
- Entity merge is reversible.
- ValueFlow class is mandatory and cannot be lost in export.
- Unknown numeric values remain unknown; null is not coerced to zero.

6.4 Testing

- Unit tests cover domain invariants.
- Integration tests cover database/storage/auth boundaries.
- Authorization tests use positive and negative cases.

- End-to-end test follows the Annex release scenario.
- Backup restore is a release test, not only an ops procedure.

7. Suggested Repository and Module Boundaries

The following structure is illustrative and SHOULD be adjusted to the selected stack without collapsing domain boundaries:

```
cs-aml/  
  backend/  
    apps/cases  
    apps/evidence  
    apps/entities  
    apps/analysis  
    apps/products  
    apps/governance  
    apps/audit  
  frontend/  
  worker/  
  infra/  
  tests/  
    unit/  
    integration/  
    e2e/  
    security/  
  docs/
```

8. Engineering Definition of Ready

- Story has SRS/feature traceability.
- Data objects and authorization scope are known.
- Acceptance criteria are testable.
- Required upstream APIs/contracts are stable or mocked.
- Security/privacy impact is identified.
- Open design decision that could invalidate implementation is resolved or explicitly time-boxed.

9. Engineering Definition of Done

- Code reviewed and merged through protected branch workflow.
- Automated unit/integration tests pass.
- Negative authorization tests pass where applicable.
- Audit/provenance behavior is verified.
- Schema/API documentation updated.
- No critical/high unresolved security defect for the capability.
- Acceptance criteria demonstrated against representative data.
- Feature traceability status updated.

10. MVP Release Evidence Package

- SRS traceability matrix showing implementation and verification status.

- End-to-end two-analyst pilot record.
- Access-control and protected-source compartment test results.
- Evidence hash, original/derivative lineage and citation tests.
- Entity merge/unmerge recovery test.
- Value-flow class persistence across DB/API/UI/export.
- Hypothesis and assessment traceability test.
- Disconfirming-search enforcement test (separate from traceability test). [v0.1.1 · A05]
- Claim/fact lifecycle test (fact creation from supporting claims/evidence, independent establishment, dispute/supersede, dependent flagging). [v0.1.1 · A10]
- Independent peer-review and dissemination-approval test.
- Backup/restore test record.
- Known limitations and formally accepted exceptions.

11. Post-MVP Deferred Work

The following capabilities remain outside the mandatory MVP unless a release dependency is discovered: automated OCR pipelines, entity NLP extraction, OpenSanctions screening, OpenAleph connector, FollowTheMoney import/export, external graph database, advanced path/network analytics, GraphSense, Flowintel interoperability, AI summarisation/assistant, partner federation, multi-tenant hosting, advanced alerting and watch queries.

Annex A — Epic Exit Checklist

Epic	Scope	Exit rule
E0	Platform Foundation & Delivery	All stories accepted; traceability updated; domain integration test passes; no unresolved critical security/data-integrity defect.
E1	Identity, Access & Protected Sources	All stories accepted; traceability updated; domain integration test passes; no unresolved critical security/data-integrity defect.
E2	Case & Investigation Workflow	All stories accepted; traceability updated; domain integration test passes; no unresolved critical security/data-integrity defect.
E3	Source, Evidence & Document Intake	All stories accepted; traceability updated; domain integration test passes; no unresolved critical security/data-integrity defect.
E4	Entity Resolution & Relationship Model	All stories accepted; traceability updated; domain integration test passes; no unresolved critical security/data-integrity defect.
E5	Timeline & Follow-the-Value	All stories accepted; traceability updated; domain integration test passes; no unresolved critical security/data-integrity defect.
E6	Typology, Hypothesis & Assessment	All stories accepted; traceability updated; domain integration test passes; no unresolved critical security/data-integrity defect.
E7	Search & Investigation Graph	All stories accepted; traceability updated; domain integration test passes; no unresolved critical security/data-integrity defect.
E8	Intelligence Product, Review & Dissemination	All stories accepted; traceability updated; domain integration test passes; no unresolved critical security/data-integrity defect.
E9	Administration, Retention & Operational Readiness	All stories accepted; traceability updated; domain integration test passes; no unresolved critical security/data-integrity defect.

Annex B — MVP End-to-End Acceptance Scenario

1. Authenticate two analysts and one independent reviewer through configured identity provider.
2. Create classified case, owner, investigation question and charter; complete required gate.

3. Register a public source and ingest original PDF evidence; compute hash and create precise extract; record a source claim and create a PROVISIONAL fact supported by it through a verification decision; have an independent reviewer establish it. *[v0.1.1 · A10]*
4. Create two entities and one organization; record aliases/identifiers; exercise candidate match and reviewed merge/unmerge on test records, each recorded as a ResolutionDecision. *[v0.1.1 · ER]*
5. Create evidence-backed relationship, ownership/control assertion and an asset.
6. Add events and render chronological timeline.
7. Create DIRECT/DOCUMENTED and RECONSTRUCTED/HYPOTHETICAL value-flow examples and verify visual/export distinction.
8. Map indicators to one typology, record counter-indicator, and create at least two competing hypotheses.
9. Create intelligence gap and confidence-rated assessment linked backward to evidence.
10. Use search and graph to retrieve only objects authorized to the case team.
11. Generate intelligence product and evidence index; submit independent peer review; revise if required.
12. Approve dissemination and create controlled export/referral package; verify sharing log and audit event.
13. Attempt unauthorized access from non-member account and confirm non-disclosing denial.
14. Run backup and restore test; verify evidence hashes and canonical references after restore.

Annex C — Initial Engineering Decision Log

Decision	Topic	Guidance	Status
ED-01	Backend framework	Select implementation stack compatible with SRS and architecture; default reference profile may use Django/DRF.	Open until kickoff
ED-02	Frontend	Select SPA/server-rendered approach; must support graph/timeline interactions and accessible forms.	Open until kickoff
ED-03	Evidence storage	S3-compatible object store vs hardened filesystem.	Decide before E0-ST03
ED-04	Search	PostgreSQL FTS for MVP vs external OpenSearch.	Recommend PostgreSQL FTS for MVP unless scale requires otherwise
ED-05	Graph	Relational/derived graph in MVP vs Neo4j/Memgraph.	Recommend derived relational/API graph for MVP; external graph post-MVP
ED-06	IAM	OIDC provider and MFA policy.	Required before E1
ED-07	Deployment	Single-organisation deployment profile for MVP.	Recommended